

# Detection Is Not Actionability: When Should LLM Query Expansion Be Applied?

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## Abstract

LLM query expansion can improve retrieval, but applying it unconditionally also harms a measurable fraction of queries. A natural response is to apply it selectively: expand only queries where retrieval is predicted to be failing. We test this directly. Qrel-free retrieval telemetry, recorded before any action, is a reasonably effective detector of retrieval failure (AUC 0.649 on TripClick TAIL test, replicating at 0.680 on an independent BEIR Natural Questions split). The same kind of telemetry, recorded after a specific expansion action, is markedly worse at predicting whether that action helped (AUC 0.552 on TripClick, 0.556 on NQ) — a statistically significant gap on both corpora. We then test the obvious consequence: does routing expansion on the easier signal alone, a failure-selective controller, recover selective-QE’s promise? It does not: the failure-selective policy’s point estimate is the best of three real policies tested, but is not significantly different from unconditional expansion, from doing nothing, or from a more expensive closed-loop agent that also consults post-action telemetry. Detecting that retrieval needs help is not the same problem as knowing what will actually help it, and the gap between the two is where selective query expansion’s promise currently stalls.

## 1 Introduction

Large language models can generate useful pseudo-documents, alternative phrasings, and disambiguating context for a search query, and corpus-grounded expansion — conditioning the rewrite on retrieved evidence rather than the query alone — improves on naive query-only rewriting. But LLM query expansion (QE) applied unconditionally can also be net-harmful on a measurable fraction of queries, particularly ones that are already well served or already underspecified in a way expansion cannot fix. Existing work establishes that QE *can* help; it leaves open when a system should trust an expansion enough to apply it, and in practice QE is deployed either unconditionally or not at all.

The natural fix is selective expansion: identify queries where retrieval is already failing, and expand only those. This requires two different predictive judgments, which prior query-performance-prediction and selective-expansion work has not clearly separated:

$$\underbrace{P(\text{retrieval failure} \mid x)}_{\text{detection}} \neq \underbrace{P(\Delta \text{nDCG} > 0 \mid x, a)}_{\text{actionability}}$$

Detection asks whether something is already wrong. Actionability asks whether a *specific* intervention will fix it. A system could answer the first question well and the second poorly, in which case selective expansion gated on detection alone would not actually recover expansion’s promise — it would route the right queries to an intervention whose benefit is still a coin flip.

**Research question.** Can retrieval telemetry that identifies poorly served queries also determine when LLM query expansion should be applied?

**Contributions. (C1) Detection  $\neq$  actionability.** Pre-action retrieval telemetry identifies failing retrieval significantly better than post-action telemetry predicts whether a specific QE action helped, replicating on an independent second corpus. **(C2) Detection alone does not solve selective QE.** A failure-selective controller built directly on the stronger signal has the best point estimate of the policies tested, but does not significantly outperform unconditional QE, BM25 alone, or a more expensive closed-loop agent. **(C3) The gap is not corpus-specific.** TripClick’s detection-vs-actionability ordering (0.649 vs. 0.552) replicates on BEIR Natural Questions (0.680 vs. 0.556), an open-domain corpus unrelated to TripClick’s clinical queries.

## 2 Related Work

**LLM query expansion.** Query2doc [10] generates pseudo-documents from an LLM and appends them to the query; MuGI [11] generates and fuses multiple expansions. Both treat expansion as unconditional, one-shot generation, conditioning on neither retrieval-state telemetry nor a harmful rewrite’s recoverability.

**Query performance prediction and selective expansion.** Classic query performance prediction [2] and selective query expansion [3] ask whether a query is hard, or likely to benefit from expansion, from retrieval signal alone — the same underlying question as our detection task. Neither separates that question from the distinct, harder one of whether a *specific* candidate action will help: prior selective-QE evaluations report whether expansion was gated well in aggregate, not whether the gating signal and the treatment-effect signal are the same predictive problem or two different ones of differing difficulty. We make that separation explicit and test it directly.

**Selective rewriting and adaptive retrieval.** Two recent, independent studies reach a conclusion consistent with ours through different mechanisms. Abe et al. [1] show LLM QE degrades retrieval specifically when the model lacks query knowledge or the query is ambiguous — evidence that unconditional expansion’s harm is systematic, not noise, reinforcing why selective gating is worth attempting in the first place. Kotte [5] tests selective gating directly: on three BEIR domains with dense retrievers and prompt-only (non-grounded) rewriting, a feature-based gate achieves AUC 0.593 and does not reliably beat never-rewriting, with an oracle ceiling of only +3pp (6% relative). This corroborates our Result 2 on an unrelated retrieval stack, rewriting mechanism, and domain set, strengthening the case that the difficulty is general rather than an artifact of either study’s setup. Neither that work nor Adaptive-RAG [4], which routes queries to a retrieval depth using a learned complexity classifier validated only on end-to-end accuracy, separately measures whether a gating signal’s own detection accuracy

matches its accuracy at predicting the chosen intervention’s benefit. We make that decomposition explicit, quantify the gap directly as two different AUCs, and show it is significant and replicates across two further corpora.

### 3 Selective Query Expansion Setup

**Datasets.** TripClick [7] is the primary benchmark: 1,523,878 documents, split into head/torso/tail query-frequency buckets, each with train/validation/test splits (1,175 queries per split per bucket); we report the TAIL split, click-derived labels. BEIR Natural Questions [6, 9] is an independent replication corpus, open-domain and unrelated to TripClick’s clinical queries, with no native train/val/test partition; we carve a disjoint, fixed-seed (42) split ourselves (800 validation) and fit fresh detectors, never reusing TripClick’s weights. An initial  $n=600$  pilot touch of test was directionally consistent with TripClick but too imprecise to resolve the AUC ordering on its own; we then ran the identical frozen pipeline, unchanged, as a single confirmatory test on the entire remaining, disjoint pool of 2,052 never-before-touched NQ queries, reported here in full and never pooled with the pilot. The  $n=2,013$  in Table 1 is this confirmatory set’s acted-upon subset.

**Retrieval failure**, the detection target, is defined as BM25 returning zero relevant documents in its top 10 results for that query — a binary, click-derived judgment available without any model predictions, fixed identically across both corpora.

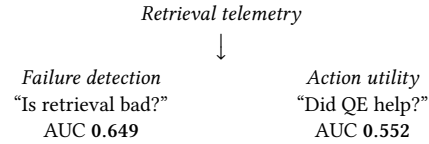
**Policies compared.** All four share the same retriever, grounding, and generator, differing only in when/whether QE fires:

| Policy               | Decision rule                                         |
|----------------------|-------------------------------------------------------|
| BM25                 | Never expand                                          |
| Static QE            | Always expand                                         |
| Failure-selective QE | Expand iff $P(\text{fail}   x) > \tau$                |
| Closed-loop QE       | Expand, then accept/rollback on post-action telemetry |

$\tau$  is the median predicted failure probability on validation, fixed before touching test, never swept or re-chosen against any reported result. Failure-selective QE uses *only* pre-action telemetry; Closed-loop QE additionally consults post-action telemetry through the same action-utility model as Result 1 (Section 4), scored against thresholds calibrated on validation only: a sufficiently positive score *accepts* the rewrite via reciprocal-rank fusion with the pre-action ranking, a sufficiently negative score *rolls back* to the pre-action ranking, and an ambiguous score *replans* with a different expansion operator, capped at two generator calls per query.

**Implementation.** Sparse retrieval is BM25 over a SQLite FTS5 index, reranked with cross-encoder /ms-marco-MiniLM-L6-v2 [8], a cross-encoder fine-tuned on MS MARCO passage ranking. Generation uses qwen2.5:3b, 4-bit quantized (Q4\_K\_M), served locally via Ollama with greedy decoding (temperature 0), capped at two calls per query episode with a 32–64 token structured output — the language model is never the entire decision surface, only one typed action within a small, fixed action space. All experiments run on a single consumer machine (Apple M4 MacBook Air, 16GB unified memory), no GPU cluster or cloud spend. All retrieval-config, detector, and threshold decisions are made on validation only. Reported confirmatory results use frozen configurations, with no subsequent

model, threshold, or retrieval decision conditioned on those outcomes; earlier implementation-debugging accesses to test, from fixing two apparatus defects mid-development, are disclosed and documented in the supplementary repository rather than omitted. Significance throughout is a paired bootstrap (10,000 resamples, seed 42). Every number in this paper is produced by a numbered, archived script in that repository and is reproducible from frozen artifacts without rerunning the generator: Table 1’s AUCs and paired differences from the frozen detection/calibration runs; Table 2 and Section 6’s oracle, selectivity-frontier, and failure-selective figures from one shared frozen checkpoint, combined under each policy’s decision rule; and Section 6’s validation-oracle robustness check from a dedicated script that touches no test data.



### 4 Result 1: Detection Is Easier Than Actionability

A failure detector — a logistic classifier on eight pre-action retrieval-telemetry features (top retrieval score, score margin, score entropy, query-term coverage, mean rare-term IDF, entity overlap, lexical result coherence, and ranking stability under a small controlled query perturbation) — is moderately predictive of whether retrieval has already failed (5-fold CV on validation, touched once on frozen test). An action-utility model, observing a *specific* candidate action’s consequences through ten post-action features (score and coverage *gains*, drift, reranker disagreement, and reranker-specific signals the detector does not have access to) fit by linear regression against the true nDCG change, is markedly weaker at predicting whether that action helped. The action-utility model has more features, including reranker signals entirely unavailable to the detector, yet still under-performs it — the gap in Table 1 is not an artifact of the detector having a richer or more favorable feature set.

**Table 1: Detection vs. actionability AUC, both corpora. Both numbers per corpus are frozen, evaluated once under the corpus’s fixed configuration.**

| Task              | TripClick | BEIR NQ | Difference |
|-------------------|-----------|---------|------------|
| Failure detection | 0.649     | 0.680   | —          |
| QE action utility | 0.552     | 0.556   | —          |
| Paired difference | +0.091*   | +0.119* |            |

\*: 95% CI excludes zero (TripClick: [+0.045, +0.136],  $n=1157$ ; NQ: [+0.076, +0.162],  $n=2013$ ).

The gap is not an artifact of comparing different noise levels: both numbers per corpus come from the same features, protocol, and population, and a paired bootstrap on the acted-upon subset confirms the difference itself, not just non-overlapping point estimates. This ordering — pre-action telemetry beats post-action telemetry at predicting intervention utility — replicates on an open-domain corpus unrelated to TripClick’s clinical queries, providing evidence that the ordering is not specific to TripClick.

## 5 Result 2: Does Detection Enable Selective QE?

The natural response to Result 1 is: detection is hard enough, fine — just expand the queries the detector flags as failing. We test this directly rather than assume it follows.

**Table 2: TripClick TAIL test ( $n=1175$ ), frozen configuration. Gain is mean nDCG@10 relative to BM25 alone; NS: 95% CI crosses zero.**

| Policy               | nDCG@10 | $\Delta$ vs. BM25 |    |
|----------------------|---------|-------------------|----|
| BM25                 | 0.3152  | —                 |    |
| Static QE            | 0.3159  | +0.0007           | NS |
| Failure-selective QE | 0.3179  | +0.0027           | NS |
| Closed-loop QE       | 0.3161  | +0.0009           | NS |

Failure-selective QE’s point estimate is the best of the three real policies — higher than unconditional Static QE and higher than the far more expensive closed-loop agent — while routing only 48.7% of queries to expansion. But none of the three is significantly different from BM25, and failure-selective QE is not significantly different from either Static QE or the closed-loop agent directly. The detector that cleanly separates failing from healthy retrieval (Result 1) does not, by itself, translate into a selective-expansion policy that reliably beats doing nothing.

Cost favors the simple policy regardless. The closed-loop agent costs 24% more generator calls (1.22 vs. 0.98 per query) and 36% higher mean latency (15.1s vs. 11.1s) than unconditional Static QE for no significant gain over it. Failure-selective QE, by construction, invokes the generator on only 48.7% of queries, versus 100% for both Static QE and the closed-loop agent — the cheapest of the three real policies tested is also the one with the best point estimate, even though none is statistically distinguishable from the others.

## 6 Why Selective QE Remains Difficult

A null policy result is ambiguous: either no benefit exists to select for, or a selector exists but ours cannot find it. A retrospective oracle — per query, the better of BM25 or Static QE under true click labels, never deployable — separates these: it gains +0.0246 nDCG@10 over BM25 ( $p < 0.001$ ), while unconditional Static QE gains only +0.0007. Static QE helps 11.6% of queries and harms 10.6%, a real, roughly balanced mix a perfect selector could exploit. At a 20% intervention quota, oracle selection reaches +0.0246; verifier-score selection reaches only +0.0025, about 10% of the oracle’s gain. This oracle is itself susceptible to selection optimism — it selects per query using the same sparse click labels it is evaluated on. Checking the identical oracle definition on TAIL *validation*, never used to select the oracle itself, shows the magnitude is split-sensitive on TripClick: +0.0533 nDCG@10 there versus +0.0246 on test, non-overlapping CIs, while on NQ the two nearly agree (+0.0399 validation vs. +0.0415 confirmatory test). We therefore read the oracle as directional evidence that heterogeneous, query-specific treatment effects exist, not as a precise, deployable bound; a fully debiased estimator (cross-fitting the selection and evaluation folds, or repeated relevance draws) is left for future work. The evidence as a whole points to *action-utility prediction*, not the absence of opportunity, as selective QE’s binding constraint.

**Why the two tasks differ in difficulty.** One plausible explanation is what each task conditions on. Detection asks a question about a state that already exists and has already been fully observed: is *this* ranking, as retrieved, adequate? Action-utility asks a counterfactual about a state that does not yet exist: would *this specific*, LLM-generated rewrite have produced a better ranking than the one already in hand? The second question’s target is itself a high-variance artifact — a single sampled generation, conditioned on retrieved evidence the generator did not choose and may misuse — so even a telemetry vocabulary rich enough to characterize the current state well need not characterize the effect of an intervention that state did not determine. This is consistent with Table 1’s asymmetry despite the action-utility model’s richer feature set, and with why gating signals in general (Section 2) struggle to double as treatment-effect predictors.

**Limitations.** The generator is a small, locally hosted 3B-parameter model under a two-call budget; a larger model might shift either AUC or the policy comparison. Corpus coverage is two domains, one clinical and one open-domain; broader coverage remains untested. Click-derived labels are an imperfect relevance proxy inherited from TripClick.

## 7 Conclusion

Detecting that retrieval has already failed is significantly easier than predicting whether a specific LLM query-expansion action will fix it, on two unrelated corpora. As a direct consequence, neither unconditional QE, a failure-selective controller built on the stronger detection signal, nor a more expensive closed-loop policy that also consults post-action telemetry significantly improves a properly-tuned retriever on ordinary queries. Selective query expansion research should therefore evaluate intervention-utility prediction directly, rather than treating query difficulty or retrieval-failure detection as a proxy for treatment benefit — the two are empirically different problems of different difficulty, and conflating them is where selective QE’s promise currently stalls.

**Experimental integrity.** All model, threshold, and retrieval configuration choices were made on validation data. Implementation defects surfaced during development were diagnosed from self-evidently implausible results and fixed before any confirmatory test evaluation; the complete experimental history, including every implementation-debugging touch, is documented in the supplementary repository rather than omitted.

## Ethical Considerations

All data used (TripClick, BEIR Natural Questions) are existing public or access-granted research benchmarks, used under their stated licenses; TripClick’s click logs are aggregate, de-identified search interactions released for non-commercial research, and we do not attempt to re-identify individual searchers. No new human-subjects data was collected. The generator is a small, locally hosted model used only for corpus-grounded query rewriting within a bounded action space, never for open-ended generation or decisions outside retrieval. Our central finding — that detecting retrieval failure is easier than predicting whether a specific intervention helps — is a caution against deploying LLM query expansion (or similar generative interventions) on the strength of a failure/difficulty detector

alone; we see this as reducing the risk of over-trusting such systems, not introducing new risk.

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